Econ 272 / MGTECON 607: Intermediate Econometrics II – RDD Overview

May 22, 2025

1 Review of RDD

1.1 Questions

1.1.1 Lecture 15: Regression Discontinuity

1. Explain the RD assumption in words and figures.
2. Describe how to implement RD in practice.
3. How can we estimate fuzzy RD by instrumental variables?
4. What supplementary analyses can be useful for RD designs?

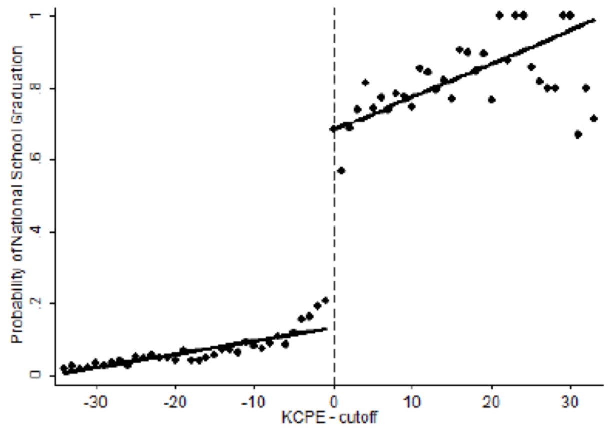
1.2 Regression Discontinuity Designs (RD)

An RD design is a great identification strategy for programs/policies/treatments that were assigned using a transparent rule with criterion based on clear cutoff values. In other words, the probability of receiving treatment is discontinuous in the covariate denoting this cutoff value. Examples here include a GPA cutoff, an annual income cutoff, etc. Comparisons of individuals that are similar, but on different sides of the cutoff point can be a credible way to estimate causal effects for a specific subpopulation: the popultion close to the cut-off. Remember that, essentially, with any observational data method we are trying to recreate an experiment and achieve "as if" randomization for a subgroup or for individuals with specific covariate values. Here, this is achieved by comparing units just above and just below the threshold as they should be pretty similar in terms other than that their value was slightly above/slightly below the cutoff value.

Note. This strategy is great for internal validity, but it does not have much external validity.

1.2.1 Basics

We distinguish between two cases of RD designs: (1) sharp regression discontinuity and (2) fuzzy regression discontinuity. We denote the cutoff point for variable X_i by c, so at $X_i = c$ the incentives to participate change. In the sharp regression discontinuity design, we assume that these incentives are strong enough to move all individuals. In the fuzzy one, we assume that this is not the case and not all individuals change their participation. X is often referred to as the forcing variable. Figure 1 gives an example for what the variable of interest looks like as a plot of the forcing variable. This is one of the first things we typically do that is different from other identification strategies: we plot the conditional mean function $\mathbb{E}[Y_i|X_i = x]$ and look for a discontinuity around the cut-off point. Oftentimes these plots will be binned histograms - make sure you don't oversmooth the plots as you might otherwise smooth out your discontinuity.



Clemens & Tiongson (2012) use a language test in the Philippines that permitted workers to travel to Korea, in order to see impacts on household investments.

Figure 1: RDD Example

1.2.2 Sharp Discontinuity (SRD)

We talk about a sharp regression discontinuity if the treatment indicator is defined as

$$W_i = \mathbf{1}_{\{X_i \geq c\}}.$$

Note that extrapolation is unavoidable here because by definition we will never get to observe both treatment and control for $X_i = c$. Hence, we must rely on extrapolation and, thus, we cannot afford to be agnostic about functional form assumptions.

Example. One example of an SRD design is Lee (2007) who was investigating the effect of incumbancy on election outcomes. They compared the election outcomes in cases where previous elections were very close.

Assumption. The key assumption for this design is that $\mathbb{E}[Y(0)|X = x]$ and $\mathbb{E}[Y(1)|X = x]$ are continuous in x.

Estimand. We define the following estimand, which is difference of the two regression functions at the cutoff point, i.e.

$$\tau_{SRD} = lim_{x \downarrow c}\mathbb{E}[Y_i|X_i = x] - lim_{x \uparrow c}\mathbb{E}[Y_i|X_i = x].$$

Estimation. We use a local linear regression approach to obtain an estimate of τ_{SRD}. For example, using a single regression, we can estimate

$$Y_i = \alpha + \beta(X_i - c) + \tau W_i + \gamma(X_i - c)W_i + \varepsilon_i,$$

which solves

$$\min_{\alpha,\beta,\tau,\gamma} \sum_{i=1}^{N} \mathbf{1}_{c-h \leq X_i \leq c+h} \times (\alpha + \beta(X_i - c) + \tau W_i + \gamma(X_i - c)W_i + \varepsilon_i)^2,$$

where h is called the bandwidth (that is, what are the cutoff points for which points close to c we include). Imbens and Kalyanaraman (2012) find that the optimal bandwidth is given by a very complicated formula that you can find in the slides :). The one important take-away here is that it is of magnitude $N^{-1/5}$.

Note. Local constant regressions (e.g. standard kernel regression, taking simple averages of units just below and just above the cut-off) have low rates of convergence as the bias is high, especially for the boundary points. It is thus not recommended for estimation. Global polynomial regression is more popular, but also not recommended since it is sensitive to the order of polynomial used.

Alternative estimation. In recent work by Armstrong and Kolesar (2018) and Imbens and Wager (2018), estimators for the average treatment effects have been redefined in terms of a weighted estimator. Let $\mu_0(x) = \mathbb{E}[Y_i|X_i = x, x \leq c]$ and $\mu_1(x) = \mathbb{E}[Y_i|X_i = x, x > c]$. Define weights $\omega_i, i = 1, ..., N, \sum_{i:X_i>c} \omega_i = \sum_{i:X_i \leq c} \omega_i = 1$. Then, the estimator is

$$\hat{\tau}(\omega) = \sum_{i:X_i>c} \omega_i Y_i - \sum_{i:X_i<c} \omega_i Y_i.$$

For any choice of $\mu_w(\cdot)$ and any set of weights, we can figure out the bias in the estimator and construct valid confidence intervals. The idea in both papers i sto choose the weights to minimize squared error, over a space of functions. There is a trade-off here when choosing the weights: when you put all the weight close to c, then the maximal bias gets small, but the variance gets large. Spreading out the weights more leads to large maximal bias, but small variance. The two papers differ in what restrictions they place on the space of functions to minimize over. AK restrict the set to satisfy Lipschitz conditions while IW restrict the set by assuming smoothness, but restricting the second derivates of $\mu_w(\cdot)$.

Picture. Figure 2 shows the assumptions, estimand and (almost all of) estimation in one picture. Here we plot the observed outcomes as a function of our forcing variable. At $X = c$, we see a discrete jump. The assumptions underlying this are drawn in blue and green - the potential outcomes are all continuous in x. The estimand here is drawn in light pink. It will the limit of the differences between the observed values right above and right below the threshold. For estimation. we typically pick an additional bandwidth (not featured here, but rather a linear approximation on all the data) and we take the difference between the estimated values at the threshold c.

1.2.3 Fuzzy Discontinuity (FRD)

We talk about a fuzzy regression discontinuity if the treatment indicator has the following property

$$lim_{x \downarrow c} P(W_i = 1|X_i = x) \neq lim_{x \uparrow c} P(W_i = 1|X_i = x).$$

Basically, here, we investigate the probability of receiving treatment as a function of X_i. If the threshold implies that we got a function of the probability that is not 0 for all values lower than c and 1 for all values above **and** it also displays a discontinuity, that is exactly what we are looking out for in this design. See Figure 3 for a visualization of this.

Example. An example of an FRD design is given in Van der Klaauw (2002) who investigated what the effect of financial aid offers was on acceptance of college admissions. The college admissions

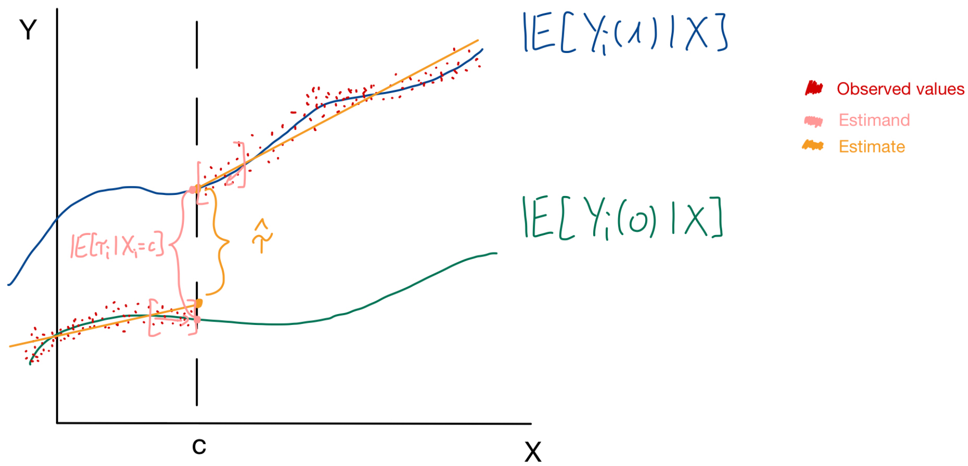


Figure 2: Sharp RD Design

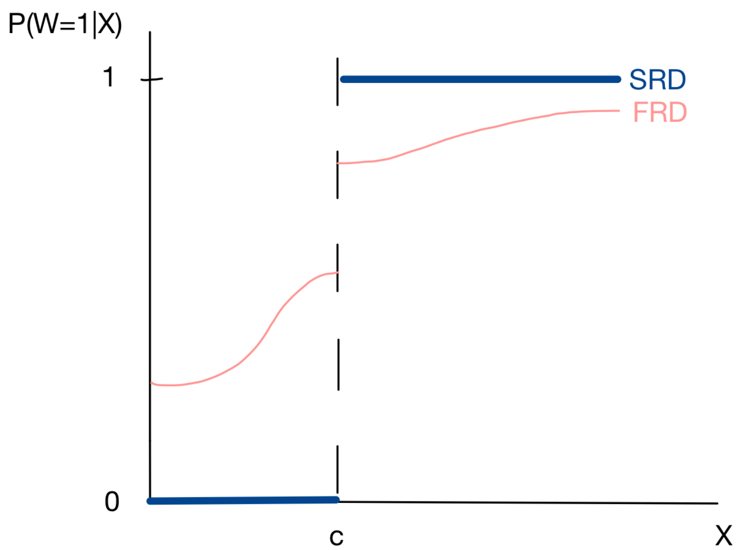


Figure 3: Propensity Score as a Function of X_i for Sharp RD Design (SRD) and Fuzzy RD Design (FRD)

office puts applicants in a few categories based on numerical scores and the financial aid offer is highly correlated with the category. They then compare the individuals close to the cutoff score.

Assumption. We first define the potential outcome $W_i(x)$ which is the potential treatment status given cutoff point x, for x in some small neighborhood around c (which requires that the cutoff point is at least in principle manipulable). This means that $W_i(x) = 1$ if unit i would receive or take up treatment if the cut-off point is equal to x. We frame out potential outcomes for the treatment indicator here in terms of moving the cutoff point because those are the different counterfactuals we want to consider. What we actually observe is that the cutoff point is $x = c$, but in our counterfactual analysis, we are interested in what would have happened if we considered other cutoff points that c. Our additional assumption for the FRD is

$$W_i(x) \text{ is non-increasing in } x \text{ at } x = c.$$

In words, this assumption is similar to the monotonicity assumption we imposed for IV, also referred to as the "no defier" assumption. If we consider a tiny change in the cutoff point to the left of c, namely $c' = c - \varepsilon$, then $W_i = 1$ at $x = c$ because c is above the threshold, assuming compliance. Similarly, if we consider a tiny change to the cutoff to the right of c, then $W_i = 0$ because c is below the threshold, assuming compliance. For the always- and never-takers, the values for W_i won't change and simply stay flat. In summary, $W_i(x)$ is assumed to be non-increasing in x at $x = c$ thus ensures that we rule out defiers. In words, we assume that crossing the threshold results in no change of treatment status, or a change of treatment status in the same direction for every unit i. In other words, when the running variable crosses the threshold, it cannot simultaneously **cause** some units to take up treatment and others to reject it. We define a complier to be a unit such that

$$lim_{x \downarrow X_i} W_i(x) = 0 \quad \& \quad lim_{x \uparrow X_i} W_i(x) = 1.$$

In words, compliers are units with running variable value X_i that would have observed treatment status being equal to 1 if the cutoff were at X_i or below (x approaching X_i from below) and they do not get the treatment if the cutoff is higher than X_i.

Estimand. In this case, we look at the ratio of discontinuities in regression functions of outcome and treatment. It has a little bit of a flavor of IV. The estimand looks as follows

$$\begin{aligned}\tau_{FRD} &= \frac{lim_{x \downarrow c}\mathbb{E}[Y_i|X_i = x] - lim_{x \uparrow c}\mathbb{E}[Y_i|X_i = x]}{lim_{x \downarrow c}\mathbb{E}[W_i|X_i = x] - lim_{x \uparrow c}\mathbb{E}[W_i|X_i = x]} \\ &= \mathbb{E}[Y_i(1) - Y_i(0)|\text{unit } i \text{ is a complier and } X_i = c].\end{aligned}$$

We interpret this ratio as the expected individual treatment effect given that unit i is a complier and $X_i = c$. In other words, we get the average effect for people at the threshold or close to it who complied with their treatment assignment.

Recommendation for practice: If you are interested in having an estimand with external validity, this is not the right choice and you might want to consider a different identification strategy. It is only valid for a population defined by the cutoff value c and by the subpopulation that is affected by it.

Estimation. In practice, we estimate this using local linear regressions for both outcome and treatment indicator

$$(\hat{\alpha}_{yl}, \hat{\beta}_{yl}) = \arg \min_{\alpha_{yl}, \beta_{yl}} \sum_{i: c-h \leq X_i < c} (Y_i - \alpha_{yl} - \beta_{yl}(X_i - c))^2$$

$$(\hat{\alpha}_{wl}, \hat{\beta}_{wl}) = \arg \min_{\alpha_{wl}, \beta_{wl}} \sum_{i: c-h \leq X_i < c} (W_i - \alpha_{wl} - \beta_{wl}(X_i - c))^2$$

for the left side of the cutoff value and, similarly, we can get $(\hat{\alpha}_{yr}, \hat{\beta}_{yr})$ and $(\hat{\alpha}_{wr}, \hat{\beta}_{wr})$ for the right-hand side. Then we find that

$$\hat{\tau}_{FRD} = \frac{\hat{\tau}_y}{\hat{\tau}_w} = \frac{\hat{\alpha}_{yr} - \hat{\alpha}_{yl}}{\hat{\alpha}_{wr} - \hat{\alpha}_{wl}}.$$

Note. We can alternatively retrieve numerically equivalent estimates by exploiting the TSLS characterization of the problem. We can use $\mathbf{1}_{\{X_i \geq c\}}$ as the excluded instrument. This instrument is highly predictive of treatment and assumed to not affect the outcome through anything but treatment. Assume we are back in the Angrist lottery example, where X_i is a random number and c the threshold for draft eligibility. In this example, the outcome of interest was the future log earnings. While the indicator is highly correlated with the probability of receiving treatment, there is no reason to believe that this indicator directly impacts the earnings, except through treatment.

Bandwidth. In the FRD design, we have two options for the choice of bandwidth: (1) calculate the optimal bandwidth separately for both regression functions and choose the smallest or (2) calculate the optimal bandwidth only for the outcome and use that for both regression functions.

Recommendation for practice: It is easier to use the same bandwidth and, to avoid bias, we use the bandwidth from the criterion for SRD design or the smallest.

Variance. We get asymptotic normality for the estimator. The complicated formula for the variance is given in the lecture slides. Note, though, that the variance estimator is equal to the robust variance for TSLS based on the subsample of observations with $c - h \leq X_i \leq c + h$.

Comparison to Unconfoundedness. Under the unconfoundedness assumption, we can rewrite our estimand as

$$\mathbb{E}[Y_i(1) - Y_i(0)|X_i = c] = \mathbb{E}[Y_i(1)|W_i = 1, X_i = c] - \mathbb{E}[Y_i(0)|W_i = 0, X_i = c].$$

The unconfoundedness strategy is based on units being comparable if their covariates are similar. In the FRD setting, we are assuming that the probability of receiving treatment is discontinuous in the covariate, thus unconfoundedness might not be the most plausible assumption here. Unit with similar values of the forcing variable (but on different sides of the threshold) must be different in some important way related to the receipt of treatment, which is problematic for unconfoundedness. Moreover, the FDR assumptions imply that comparing treated and control units with $X_i = c$ is the wrong approach as treated units with $X_i = c$ include compliers and always-takers and control units at $X_i = c$ are never-takers. However, even if unconfoundedness holds, under continuity of potential outcome regression functions, the FRD approach is consistent for the ATE for the complier at $X_i = c$. An analysis based on unconfoundedness can only be credible if there is a substantive argument that the difference in the forcing variable is immaterial for the comparison of the outcomes of interest.Note that nothing in the data will tell you that unconfoundedness would be the wrong thing to do (except potentially a discontinuity in the propensity score around $X_i = c$), but leaving the extra knowledge we have about te cut-off point on the table and not exploiting that feels insufficient.

1.2.4 Concerns about Validity

There are two main conceptual concerns in applying RD designs - for both sharp and fuzzy designs:

1. **Other changes:** possibility of other changes at the same cutoff value of the covariate that may, for example, affect the outcome
2. **Manipulation of Forcing Variable**

1.2.5 Specification Checks

When using the RD design, there are a few robustness checks we can do to validate our use of this method:

1. Discontinuities in average covariates
2. Discontinuity in distribution of the forcing variable
3. Discontinuities in average outcomes at other values
4. Sensitivity to bandwidth choice

We will go over each of these in detail below. Again, an important note here is not to oversmooth the graphs in order to be able to detect important discontinuities.

1.2.6 Discontinuities in Average Covariates

This robustness check is equivalent to testing a null effect of a zero average effect on pseudo outcomes not to be affected by the treatment. In most cases, the reason for the discontinuity in the probability of treatment does not suggest a discontinuity in the average vale of covariates. Thus, if we find such a discontinuity, it casts doubt on the assumptions underlying the RD design.

1.2.7 Discontinuity in Distribution of the Forcing Variable

This robustness check test the null hypothesis of continuity of the density of the forcing variable X against the alternative of a jump in the density at that point. This is not required by the design, but a discontinuity could suggest violations of the no-manipulation assumption.

1.2.8 Discontinuities in Average Outcomes at Other Values

This robustness check suggest to take a subsample with $X_i < c$ and one with $X_i \geq c$. In each subsample, test for a jump in the conditional mean of the outcome at the median of the forcing variable.

1.2.9 Sensitivity to Bandwidth

The last robustness check one can run is investigating the sensitivity of the inferences to the bandwidth choice. Re-run the RD for a bandwidth that is, for example, half the size or twice the size of the originally chosen bandwidth. If the results are critically dependent on a particular bandwidth choice, they are less credible than if the drawn conclusions don't change. Note that, of course, the point estimates are going to change. We are more concerned about the qualitative conclusions here.